

SYMBOLIC MICROPROGRAM

- Control Storage: 128 20-bit words
- The first 64 words: Routines for the 16 machine instructions
- The last 64 words: Used for other purpose (e.g., fetch routine and other subroutines)
- Mapping: OP-code XXXX into 0XXXX00, the first address for the 16 routines are 0(0 0000 00), 4(0 0001 00), 8, 12, 16, 20, ..., 60

Partial Symbolic Microprogram

Label	Microops	CD	BR	AD
ADD:	ORG 0			
	NOP	I	CALL	INDRCT
	READ	U	JMP	NEXT
	ADD	U	JMP	FETCH
BRANCH:	ORG 4			
	NOP	S	JMP	OVER
	NOP	U	JMP	FETCH
OVER:	NOP	I	CALL	INDRCT
	ARTPC	U	JMP	FETCH
STORE:	ORG 8			
	NOP	I	CALL	INDRCT
	ACTDR	U	JMP	NEXT
	WRITE	U	JMP	FETCH
EXCHANGE:	ORG 12			
	NOP	I	CALL	INDRCT
	READ	U	JMP	NEXT
	ACTDR, DRTAC	U	JMP	NEXT
	WRITE	U	JMP	FETCH
FETCH:	ORG 64			
	PCTAR	U	JMP	NEXT
	READ, INCPC	U	JMP	NEXT
	DRTAR	U	MAP	
INDRCT:	READ	U	JMP	NEXT
	DRTAR	U	RET	

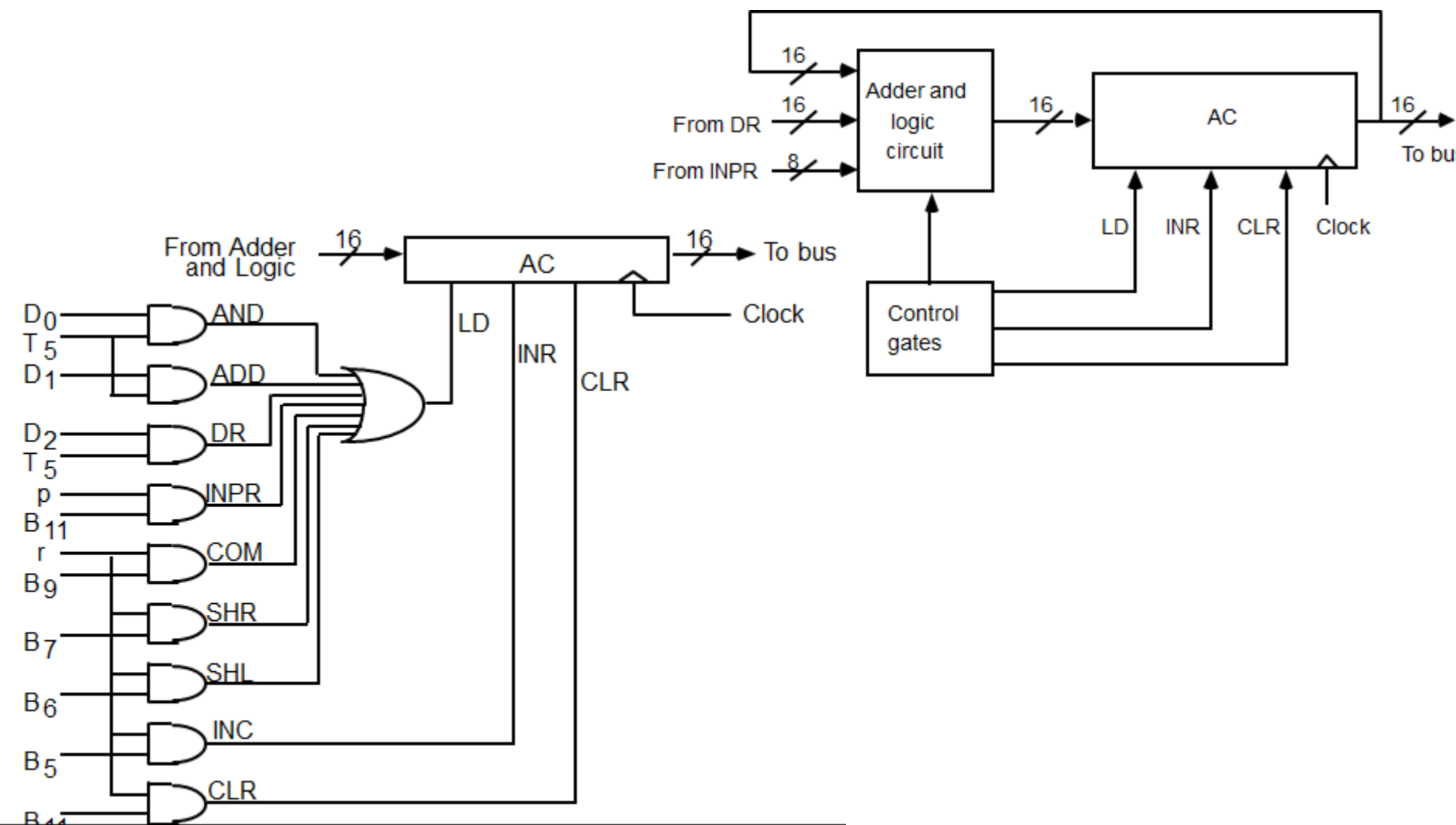


BINARY MICROPROGRAM

Micro Routine	Address		Binary Microinstruction					
	Decimal	Binary	F1	F2	F3	CD	BR	AD
ADD	0	0000000	000	000	000	01	01	1000011
	1	0000001	000	100	000	00	00	0000010
	2	0000010	001	000	000	00	00	1000000
	3	0000011	000	000	000	00	00	1000000
BRANCH	4	0000100	000	000	000	10	00	0000110
	5	0000101	000	000	000	00	00	1000000
	6	0000110	000	000	000	01	01	1000011
	7	0000111	000	000	110	00	00	1000000
STORE	8	0001000	000	000	000	01	01	1000011
	9	0001001	000	101	000	00	00	0001010
	10	0001010	111	000	000	00	00	1000000
	11	0001011	000	000	000	00	00	1000000
EXCHANGE	12	0001100	000	000	000	01	01	1000011
	13	0001101	001	000	000	00	00	0001110
	14	0001110	100	101	000	00	00	0001111
	15	0001111	111	000	000	00	00	1000000
FETCH	64	1000000	110	000	000	00	00	1000001
	65	1000001	000	100	101	00	00	1000010
	66	1000010	101	000	000	00	11	0000000
	67	1000011	000	100	000	00	00	1000100
INDRCT	68	1000100	101	000	000	00	10	0000000

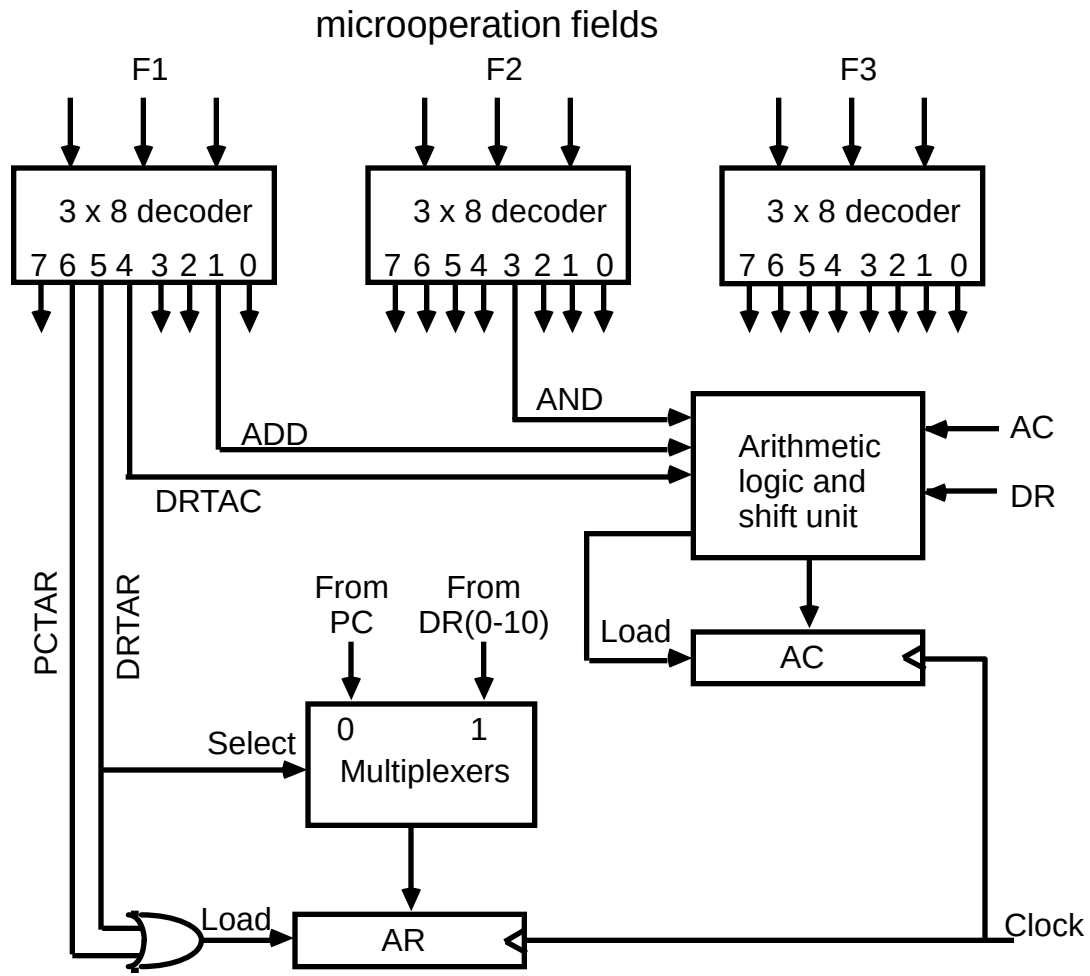
This microprogram can be implemented using ROM





DESIGN OF CONTROL UNIT

- DECODING ALU CONTROL INFORMATION -



Decoding of Microoperation Fields